Implementation of Nonlinear Classifiers for Adaptive Autoregressive EEG Features Classification

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***Abstract*—The objective of this work is to realize two nonlinear classifiers for the adaptive autoregressive Electroencephalography (EEG) features. The EEG features are modeled as adaptive autoregressive model and estimated using recurring least square algorithm. Nonlinear classification is performed using multilayer perceptron (MLP) and radial basal function neural network to classify extracted features for a two classes experiment. For validation, hands movement imaginations based experiments are conducted using low price EEG EPOC headset. A comparative study, carried out amongst the nonlinear classifiers and with a linear discriminant analysis, demonstrates the dominance of the MLP as nonlinear classifier.**

Keywords—Adaptive Autoregessive (AAR), Brain Computer Interface (BCI), Electroencephalogram (EEG), Recurring Least Square (RLS), Linear Discriminant analysis (LDA), Multilayers Perceptron (MLP), Radial Basal Fuction Neural Network (RBFN), Sensorimotor Rhythms (SMR), Emotiv EPOC headset.

I. INTRODUCTION

Recent progress in neuroscience have led to interact directly with a human brain for assistant of both disabled and healthy persons. There are variety of brain sensing techniques but usually these techniques are divided into two groups, invasive and non-invasive techniques. In the invasive techniques sensors (electrodes) are implanted directly into the major parts of brain or on the surface of the brain beneath the skull, such as Electrocorticography (ECoG) and Intracranial Electroencephalography (iEEG).These techniques are not appropriate for healthy persons because may damage brain during surgery but they have superior spatial resolution and high signal to noise ratio [1].

The non-invasive technique electroencephalography (EEG) reads brain electrical signal by placing the electrodes on scalp. Other non-invasive techniques, Magnetoencephalography (MEG) measure the magnetic fields produced by naturally occurring brain currents, functional near infrared spectroscopy (fNIRS) monitor the blood oxygenation in the form of changes in near-infrared light and functional magnetic resonance imaging (fMRI) measure the changes in the magnetic properties of hemoglobin as it carries oxygen [2].

MEG, fNIRS and fMRI have expensive and non-mobile bulky equipment as compare to EEG. Therefore, EEG is preferred due to its high temporal resolution, portable, light weight and low cost headset. The EEG can track the high rate complex patterns of brain signals as fNIRS and fMRI has low speed to capture brain signals [3].

Brain computer interface (BCI) is upcoming technology that provide a pathway to control different devices using a brain signals or thoughts. Different thoughts produce different signals in particular location of a brain, BCI infer these thoughts and processing of that thoughts with different techniques such as EEG, MEG, fNIRS and fMRI. EEG based BCI system play vital role to detect of illnesses of a human brain like epilepsy (repetitive brain seizures), tumor, stroke, drug's effects, and memory weakening, sleep sickness, head injury and growth of a human brain etc [3]. It is a helpful technology for paralyzed people to control prosthesis, peripheral devices and wheelchair. BCI has also values application for healthy people, like brain state sensing (such as of a driver during driving and of a pilot during flight, etc.) and controlling different systems (such as remote controlling in battle filed or space mission, smart homing, and smart gaming etc.).

In modern BCI systems, there are several implementation approaches based experimental procedure such as sensorimotor rhythms (SMR) [4], auditory evoked potentials, mental activity, cognitive potential and visual evoked potential (VEP) like steady state VEP (SSVEP), time modulated VEP (t-VEP), and transient VEP (TVEP) [5].

In general the framework of a BCI system has five steps data acquisition, preprocessing, feature extraction, classification and controlling devices. Both feature extraction and classification are critical step of in the design of BCI system. Common feature extraction methods are spatial filtering, power spectral density (PSD), auto regressive (AR) model, adaptive auto regressive (AAR) model, inverse model-based features, common spatial patterns (CSP) and principal component analysis (PCA) etc [6]. The classifiers are linear and nonlinear, the linear classifiers such as linear discriminant (LDA), hierarchical discriminant component analysis (HDCA), logistic regression (LR) & linear support vector machine (SVM) and nonlinear are Bayes quadratic, artificial neural network, support vector machine, k neighbor and hidden Markov model [7].

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In this work the adaptive autoregressive (AAR) model is developed as model of brain activity, AAR model is pure mathematical model so it is appropriate for EEG analysis. The recurring least squares (RLS) estimator with linear discriminant analysis (LDA) for AAR EEG processing had been applied in paper [9]. In this paper, two nonlinear classifiers as multilayer perceptron (MLP) and radial basal function neural network (RBFN) are proposed to classify RLS based the AAR EEG features. The validation of the proposed schemes are supported by performing right and left sensorimotor imagination experiments. Previously published sensorimotor imagination dataset is used for validation of our experiments and EEG signal processing. The results show that nonlinear classifier MLP is better choice than LDA and RBFN for classification of RLS based the AAR EEG features.

II. Material and Method

A. EEG as Adaptive Autoregressive

The model of EEG as adaptive autoregressive (AAR) having order p is described as

$$y(k) = a_1(k)*y(k\text{-}1) + a_2(k)*y(k\text{-}2)\ldots\ldots a_p(k)*y(k\text{-}p) \tag{1}$$

The EEG time series and adaptive coefficients of AAR are denoted by y(k-1)….y(k-p) and a1(k)….ap(k) respectively. The order of AAR model is denoted by p. The vector form of adaptive autoregressive coefficients and prior p samples of EEG signal are described as [8]

$$\boldsymbol{A}(k) = [\ a_1(k)\ \ a_2(k)\ \ a_3(k)\ \ldots\ldots\ a_p(k)\ ]^{\mathrm{T}} \tag{2}$$

$$\mathrm{Y(k) = [\ y(k)\ \ y(k\text{-}1)\ \ y(k\text{-}2)\ldots\ldots\ y(k\text{-}p)\ ]^{T}} \tag{3}$$

where T is superscript for the transpose of vector. The estimation of error is described as

$$e(k) = y(k) - \boldsymbol{A}(k)^{\mathrm{T}} * \boldsymbol{Y}(k) \tag{4}$$

where the prediction error is e(k).

B. Feature extraction (RLS estimation)

The RLS estimation algorithm can be defined using the following interpretations

$$\boldsymbol{Z}(k) = (1+uc)*\boldsymbol{P}(k\text{-}1)\ \boldsymbol{Y}(k) \tag{5}$$

$$\boldsymbol{G}(k) = \boldsymbol{Z}(k)\ \ /\ [\mathbf{1} + \boldsymbol{Y}(k)^{\mathrm{T}} * \boldsymbol{Z}(k)\ ] \tag{6}$$

$$\boldsymbol{A}(k) = \boldsymbol{A}(k\text{-}1) + e(k)*G(k) \tag{7}$$

$$\boldsymbol{P}(k) = \boldsymbol{P}(k\text{-}1)*(1 + uc) - \boldsymbol{G}(k)*\boldsymbol{Z}(k)^{\mathrm{T}} \tag{8}$$

where G is kalman gain vector and P represent inverse of the correlation matrix. In this work, the order of AAR model is p = 6 and an update coefficient is uc = 0. Initially, the AAR coefficient A(k) are taken from old data set of EEG.

At each time instants, the AAR coefficients are estimated and generated a single vector D by stacking the AAR coefficients of each electrode as

$$\boldsymbol{D}(k) = [\boldsymbol{A}(k)^{\mathrm{Ch1}}\ ;\ \boldsymbol{A}(k)^{\mathrm{Ch2}}\ ;\ldots\ldots\ \boldsymbol{A}(k)^{\mathrm{ChN}}] \tag{13}$$

where A(k)Ch1 , A(k)Ch2 ,……. A(k)ChN characterized the estimated coefficients of EEG signal acquired by electrodes Ch1, Ch2, …… ChN respectively. In this study, two channels are used to produce a single vector D. The distance Dis is computed by above mentioned classification method using a single vector D as input vector. After this, the down sampling of distance Dis(k) is accomplished by computing average of 32 distance Dis(k) samples. The distance Dis(k) is classified as right hand movement imagination when distance Dis(k)'s value is greater than 0 and as left hand movement imagination when distance Dis(k)'s value is smaller than zero.

C. Classification

The feature vector associated to the right and left hand movement imaginations is calculated by RLS estimation, Eq. (5)-(8) and is classified by three classifiers (LDA, MLP and RBFN).

1) Linear Discriminant Analysis

Linear discriminant analysis (LDA) use linear functions to differentiate classes by generating hyper planes that separate the data set representing the different classes. The AAR coefficients are linearly combined with weight vector w linearly using linear LDA. The distance **Dis** is computed as

$$\boldsymbol{Dis}(k) = \boldsymbol{w} - \boldsymbol{D}(k)*\boldsymbol{w_o} \tag{14}$$

where weight vector is denoted by w and computed by LDA using old data set of EEG.

2) Multilayer Perceptron

A multilayer perceptron (MLP) is an artificial feed forward neural network that maps a set of input data onto a set of appropriate output(s). Feed forward networks often have one or more hidden layers of sigmoid neurons followed by an output layer of linear neurons [9]. Fig.1 show the structure of an MLP.

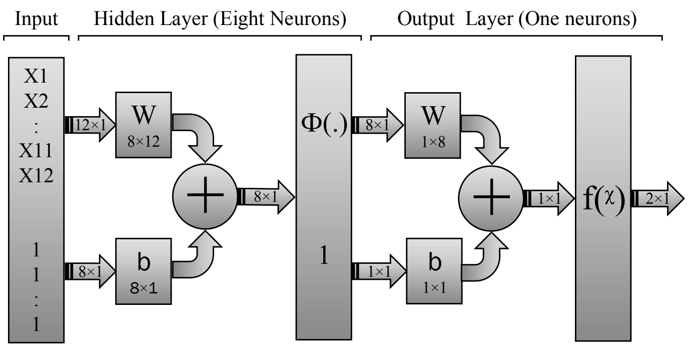

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Fig. 1. Structure of Multilayer Perceptron

In this work multilayer perceptron has 12 input neurons according to the number of channels and order (p) of AAR model, one hidden layer of eight neurons and one output neuron. The output is computed in the form of distance Dis.

3) Radial Basal Function Neural Network

Radial Basis Function neural network (RBFN) is a feed forward neural network in which radial basis function (RBF) is used as activation function. It has three layers: an input layer, a hidden layer having RBF activation function and an output layer having linear activation function [10]. Fig.2 show the structure of RBFN. In the current case, the RBFN has 12 different inputs, a hidden layer of maximum 100 neurons with spread function of value 1 and one output neuron. The output is computed in form distance Dis.

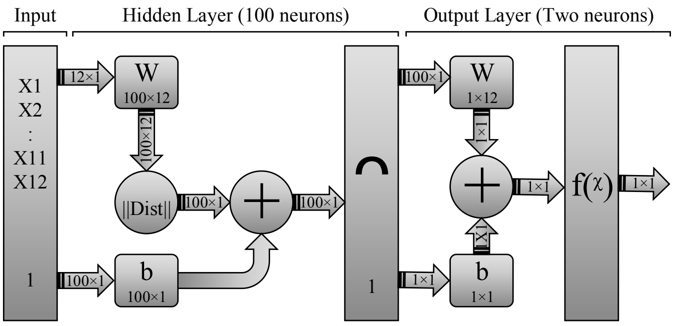

Fig. 2. Structure of RBFN

D. *Experimental Paradigm*

The E-prime (Psychology Software Tools) is used to design the time base experimental paradigm with the time accuracy of 10-3 second. The thirty six trials are represented to object during experiment with every trial duration is 10 seconds. The time line of a trial is shown in Fig.3. Each trial initiate with the representation of a fixation sign for 4 seconds. At 4 second of each trial, arrow appear for one second as instruct sign crossponding to direction of right and left hand movement imagination. The subject is instructed to imagine the right and left hand movement according to direction of arrow and the screen will remain blank during last five second [11].

E. *Data Acquisition and Preprocessing*

To get access sensorimotor cortex, the EPOC headset is placed on human head at 20 degree rotated to backward as compare to regular setting. This arrangement allow to read 10 EEG electrodes of C3, C4, CP5, CP6, FC3, FC4, T7, T8, P7 and P8 [12]. The positioning of 20 degree rotated headset and positions of electrodes are shown in Fig.4. The five healthy persons (all males having average age 23 years) are participated in hands movement imagination experiment. During the experiment, the EEG signals are recorded from two electrodes of sensorimotor cortex (C3 and C4) and sampled at 128 Hz. Inherently the EPOC headset has a DC offset in EEG signals for wireless transmission. Therefore EEG signal is filtered using high pass filter of 2 Hz cutoff frequency to eliminate the DC offset and low frequency noise. These data sets are denoted by “A” which are recorded in PIEAS.

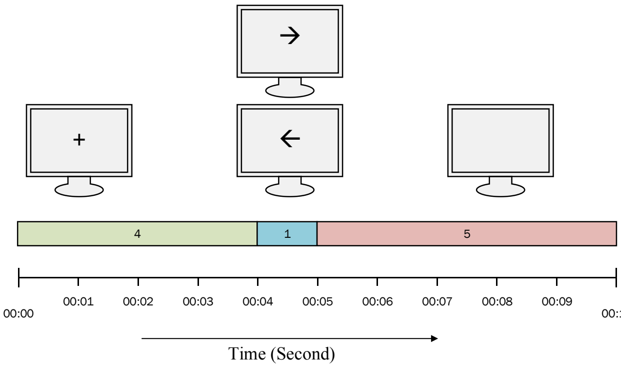

Fig. 3. Hand Movement Imagination Experimental Paradigm

The other data set (denoted by “B”) is obtained from university of technology Graz. This data is publically available and acquired from healthy subject (female, 25 years). The experimental task is same as described for data set “A” but there is little difference in experimental paradigm that is feedback bar appearing during the last five second of each trial. This data set “B” has 140 trials and recoded using G.tec amplifier having Ag/AgCl electrodes from sensorimotor cortex (C3, C4 and Cz). It is sampled at 128 Hz frequency and filtered with band pass filter (0.5 Hz to 30 Hz) by data provider [13]. Again, it is filtered using high pass filter (2 Hz) to ensure same preprocessing essentials that is performed on data set “A”.The 50% data of each dataset is used to train the classifier.

Fig. 4. 20 Degree Rotated Arrangement of EPOC Headset

III. RESULTS

The performance of each scheme is evaluated on basis the correct-trial classification (%CTC) accuracy, correct-instant k classification (%CIC) accuracy, maximum value of %CIC and average of accuracy (Avg). The main evaluation parameter is average of accuracy (Avg) because it is more general. Using LDA, MLP and RBFN classifier, the plots of %CTC accuracy are shown in Fig. 5, Fig. 6 and Fig. 7, respectively. The other evaluating parameter %CTC accuracy response is shown in Fig. 8, Fig. 9 and Fig. 10, respectively. The Avg accuracy of MLP classifier appear high in Fig. 6 and Fig. 9.

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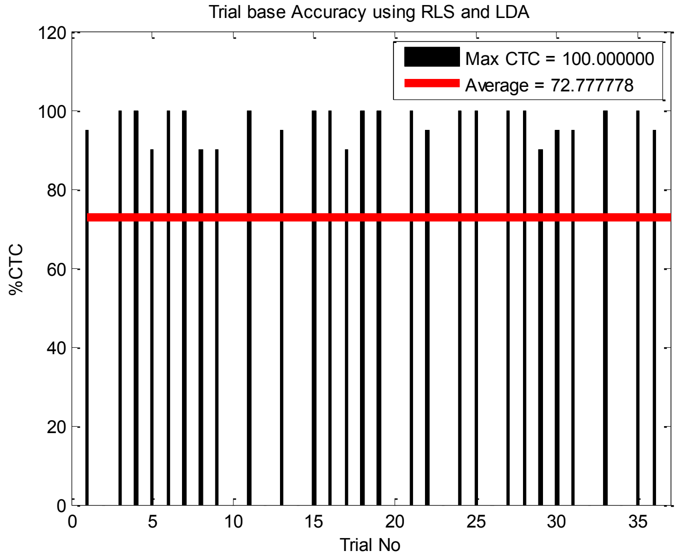

Fig. 5. Correct-Trial Classification (% CTC) accuracy by LDA

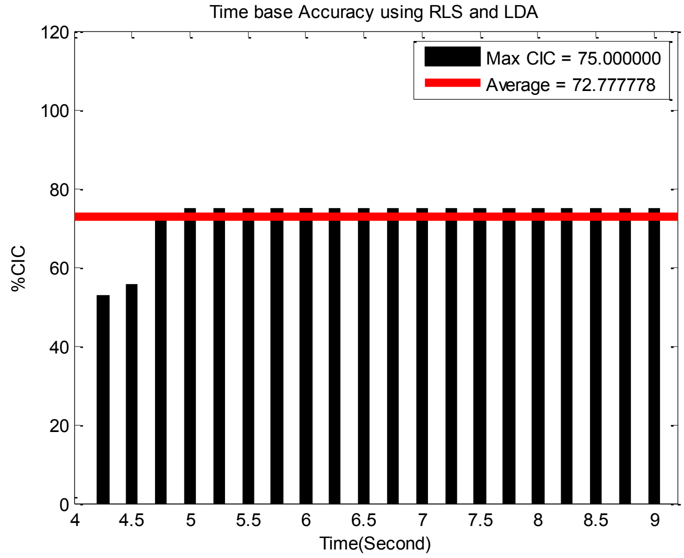

Fig. 8. Correct- Instant classification (% CIC) accuracy by LDA

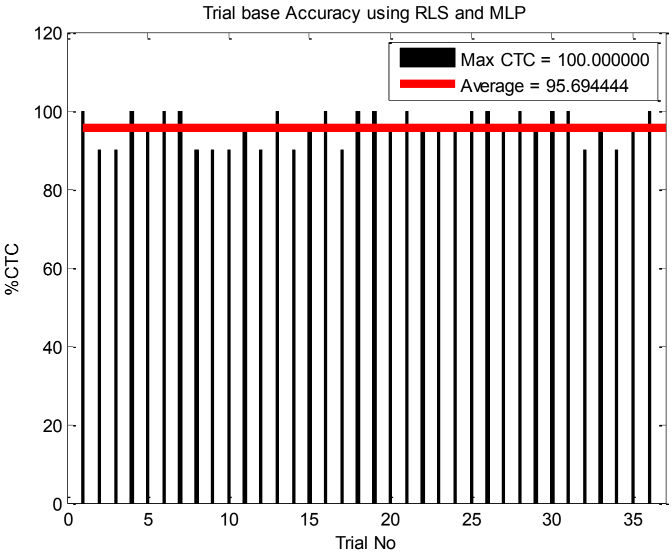

Fig. 6. Correct-Trial Classification (% CTC) accuracy by MLP

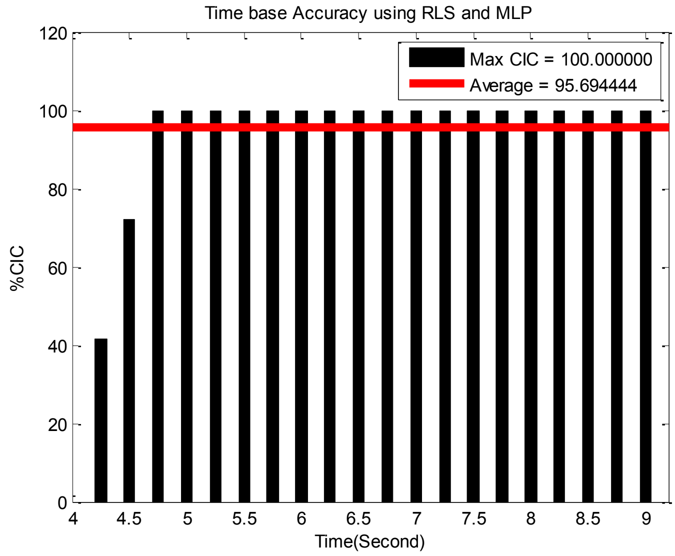

Fig. 9. Correct-Instant classification (% CIC) accuracy by MLP

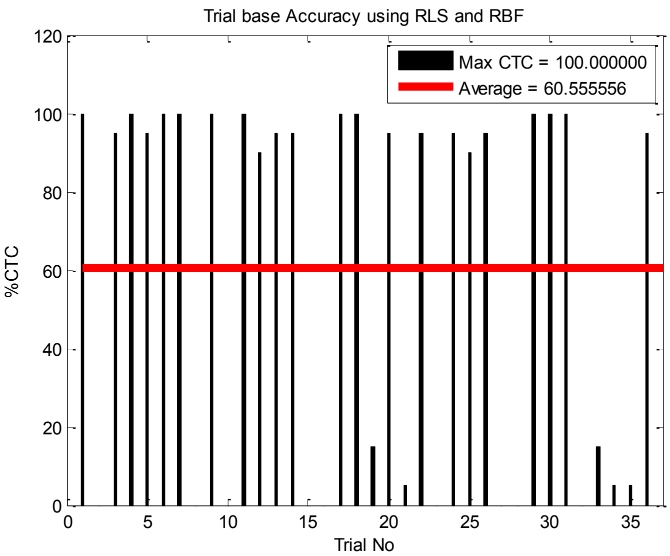

Fig. 7. Correct-Trial Classification (% CTC) accuracy by RBFN

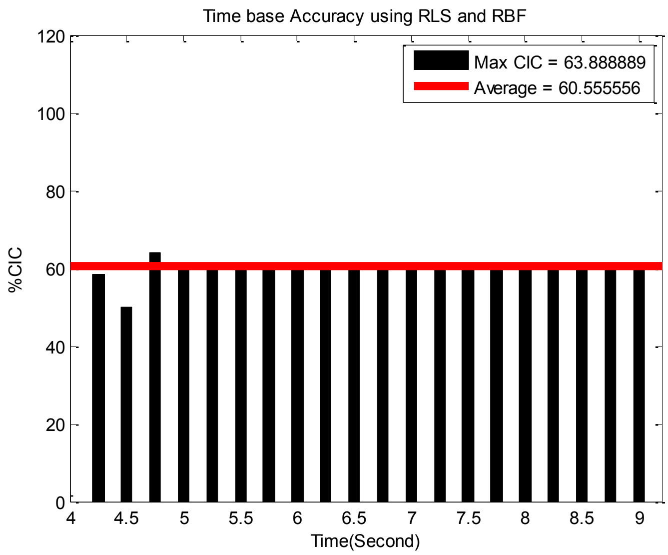

Fig. 10. Correct-Instant Classification (% CIC) accuracy by RBFN

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The Table1 shows the comparison of maximum values of %CIC and average of accuracy for each dataset. Both maximum and average values are significantly higher using nonlinear classifier MLP but it is more sensitive to training hence its classification accuracy has a range instead of single value. In Table 1, other nonlinear classifier RBFN give poor result for classification of AAR coefficients of EEG.

TABLE I. COMPARISON OF LDA,MLP AND RBFN

Data Set		Algorithms					
		LDA		**MLP**		**RBFN**	
		Max	**Avg**	**Max**	**Avg**	**Max**	**Avg**
A.1	PIEAS BCI	83.33	77.64	90±4	87±4	58.33	57.64
A.2		72.22	64.44	90±4	85±4	52.78	50.56
A.3		75.00	72.78	94±4	84±4	63.89	60.56
A.4		66.67	63.33	88±4	89±4	55.56	54.17
A.5		61.11	60.42	90±4	85±4	52.78	47.78
B	Graz BCI	87.86	78.82	90±4	88±4	95.00	87.43

IV. CONCLUSION

This paper presents the implementation of three classifiers (MLP, LDA and RBFN) for classification of EEG features modeled as AAR coefficients. The coefficients are extracted by RLS algorithm followed by their classification using the three methods. The comparative validation, carried out conducting motor imaginary experiments, demonstrates the superior potential of the nonlinear MLP classifier than the LDA and RBFN classifiers.

V. REFERENCES

[1] "Minimal Invasive Environment-Brain Interface," Jang's Lab, 2011. [Online]. Available: http://jangslab.hanyang.ac.kr/01_research/research0201.htm. [Accessed 08 08 2015].

[2] Digiora, "BCI - Brain Computer Interface," Mepits, 2015. [Online]. Available: https://www.mepits.com/tutorial/173/Electronics-Devices/BCI---Brain-Computer-Interface. [Accessed 02 09 2015].

[3] M. Telpan, "Fundanmentals of EEG Measurement," Measurement Science Review, vol. 2, 2002.

[4] C. Neuper, A. Schlogl and G. Plurtscheller, "Enhancement of Left-Right Sensorimotor EEG Differences During Feedback-Regulated Motor Imagery," Journal of Clinical Neurophysiology, vol. 16(4), no. July 1999, pp. 373-382, 1999.

[5] D. Liparas, S. Dimitriadis, N. Laskaris, A. Tzelepi, K. Charalambous and L. Angelis, "Exploiting the temporal patterning of transient VEP signals: A statistical single-trial methodolgy with implications to brain-computer interfaces(BCIs)," Journal of Neuroscience Methods, no. 30 April 2014, pp. 189-198, 2014.

[6] "Current Practices in Electroencephalogram based Brain Computer Interfaces (Information Science)," what-when-how.com, [Online]. Available: http://what-when-how.com/information-science-and-technology/current-practices-in-electroencephalogram-based-brain-computer-interfaces-information-science/. [Accessed 10 09 2015].

[7] F. Lotte, M. Congedo, A. Lecuyer, F. Lamarche and B. Arnaldi, "A review of classification algorithms for EEG based brain-computer interfaces," Journal of Neural Engineering, 2007.

[8] G. Pfurtscheller, C. Neuper, A. Schlogl and K. Lugger, "Separability of EEG Signals Recorded During Right and Left Motor Imagery Using Adaptive Autoregressive Parameters," IEEE Transactions On Rehabilitation Engineering, vol. 6, pp. 316-325, 1998.

[9] E. B. Baum, "On the Capabilities of Multilayer Perceptrons," Journal of Complexity, vol. 4, p. 193–215, 1988.

[10] A. P. Engelbrecht, Computational Intelligence An Introduction, Chichester: John Wiley & Sons Ltd, 2007.

[11] M. Ahmad and M. Aqil, "QR-RLS Adaptation of Autoregressive EEG Features," in International Conference on Intelligent Systems Engineering, Islamabad, 2016. (submitted manuscript)

[12] "Forum » Emotiv Main Forum » Emotiv SDK," Emotiv, 2014. [Online]. Available: https://emotiv.com/forum/forum15/topic1338/messages/. [Accessed 01 June 2015].

[13] A. schlogl and G. Pfurtscheller, "Data Set: BCI experiment," University of Technology Graz, Graz, alois.schloegl@tugraz.at.

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